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Place oriented human settlement value assessment in the world heritage Trang An landscape complex

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Abstract

This study evaluates and explores the impact of heritage designation on human settlement values within the Trang An Landscape Complex, a UNESCO World Heritage site in Ninh Binh province, Vietnam. Employing a place-oriented hedonic price model alongside spatially sensitive methods such as Geographically Weighted Regression and Monte Carlo simulation, the study evaluates how heritage status influences property values across core, buffer, and surrounding zones of the complex. Through a comprehensive survey of 307 households, the analysis incorporates structural attributes, locational factors, human perceptions, and mobility-related variables, reflecting both tangible and intangible influences on settlement value. The total human settlement value of the Trang An landscape complex is approximately 95.04 billion USD. Findings indicate that properties in proximity to cultural and natural landmarks exhibit higher values, emphasizing the economic importance of heritage preservation in supporting local economies. Furthermore, perception variables related to community cohesion, education quality, and environmental health demonstrate significant impacts, highlighting residents' appreciation for cultural and environmental quality in heritage zones. Ultimately, this research underscores the role of cultural and natural assets in shaping settlement values and advocates for inclusive policies that engage local communities in heritage management to promote resilience and long-term value preservation.

Keywords Human settlement value, Place-oriented hedonic price model, Trang An landscape complex, UNESCO world heritage site, Ninh Binh, Vietnam

1 Introduction

Human settlement refers to the totality of human communities — cities, towns, or villages — encompassing physical structures, social systems, and cultural practices that sustain life. “Human settlements are the territorial arrangements made by Anthropos for his own benefit and welfare. They are the results of human action and their goal is human survival, an easier and better life (especially in early childhood); happiness and safety (as Aristotle demanded); and opportunities for human development.” [4]. Settlements



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become heritage when their value is recognized through research and evaluation [5–7]. For UNESCO designation, a settlement must meet criterion (v), representing an outstanding example of traditional land-use or human interaction with the environment, particularly under conditions of irreversible change. In Vietnam, the Trang An Landscape Complex (Ninh Binh province) was inscribed under this criterion as a living example of continuous human settlement in a dynamic karst environment, reflecting human resilience and adaptation to environmental transformations.

The Trang An Landscape Complex is located on the southern edge of the Red River Delta, Vietnam [8]. Covering a total area of 12,252 hectares, the complex is divided into a core zone and a buffer zone [9]. The core heritage zone spans 6226 hectares and encompasses areas recognized nationally for their historical, geological, geomorphological, and ecological value [10]. This zone protects a range of natural features such as limestone mountains, forests, rivers, and open spaces, contributing to its heritage value [11]. Trang An is globally renowned for its striking Karst landscape, characterized by towering limestone cliffs and a vast network of water caves spanning over 30 km. These natural features contribute to the region's rich biodiversity and unique ecosystems.

Additionally, Trang An holds excellent archaeological and historical significance, with evidence of continuous human occupation for over 30,000 years, showcasing the resilience and adaptability of ancient societies to dramatic climate changes. The complex also includes key cultural heritage sites, such as the ancient Hoa Lu Citadel, Vietnam's first feudal capital, and the Bai Dinh Pagoda, adding spiritual and historical depth to the site. Recognized as a World Heritage Site in 2014, Trang An is a prime example of the interaction between humans and their natural environment, making it a site of cultural and natural importance to Vietnam [10].

Current studies on the economic valuation of human settlement reveals a theoretical trajectory moving from traditional instrumental methodologies toward distinct, place-based developmental strategies [12, 13]. The mainstream instrumental approach relies heavily on monetary tools such as Cost–Benefit Analysis and ecosystem services valuations to assess spatial interventions [14, 15]. In an effort to reconcile economic growth with societal needs, the "shared value" framework has emerged within economic geography, arguing that economic competitiveness is inextricably linked to the social conditions of the specific locations and industry clusters where productions operate [16]. Expanding further into place-specificity, the endogenous development perspective focuses on "adding value to place" by leveraging local distinctiveness—such as cultural heritage, craftsmanship, and the notion of *terroir*—as the primary basis for economic activity [17, 18]. This approach prioritizes the mobilization of internal resources and human capacities to foster economic resilience, aiming to retain financial gains within the community and transform external global pressures into self-constructed, sustainable local models [19].

International research on human settlement heritage reveals a correlation between the benefits of global recognition and the erosion of local socio-cultural fabrics, prompting a theoretical shift from static, expert-led conservation to dynamic, community-centric frameworks. Case studies from East Asia demonstrate that World Heritage Site designation often accelerates a shift from traditional agrarian industries to tourism-dependent economies; in both Ogimachi (Japan) [20] and Hahoe Village (Korea) [21], this transition has generated economic revenue and enhanced local pride but simultaneously degraded

the physical quality of life through overcrowding, noise, and severe privacy invasions that commodify residents' daily lives. This phenomenon often weakens traditional cooperative systems, such as the *yui* in Japan or clan-based structures in Korea, and fosters a "doughnut phenomenon" where original residents are displaced by commercial operators or absentee landlords [20]. In the Chinese context, this displacement is frequently exacerbated by state-led regeneration projects that physically sever the bonds between long-term residents and their historic environments [12, 22]. To address these challenges, scholars argue that traditional "value-based" assessments, which privilege expert-defined "social value," are insufficient for capturing the fluid, lived experience of place [13]. Instead, recent scholarship advocates for a "place-oriented" approach that integrates symbolic and intentional values or a "glocal community" framework [13, 23]; the latter acknowledges that in an era of overtourism and displacement, the "community of interest" may no longer physically reside at the site, necessitating digital tools and social media to maintain the continuity of emotional connections and collective memories across geographical boundaries [24].

Previous studies demonstrated that residential areas with heritage-listed properties attract higher property values than non-listed areas. A study in Ontario, Canada, found a positive correlation between heritage designation and market performance; 74% of designated properties met or exceeded average market levels, matched or surpassed general sales rates, and showed greater resilience against downturns [25]. Similarly, a study on Ku-ring-gai in Sydney's North Shore revealed that heritage-listed properties commanded significantly higher prices than non-listed ones, with listed houses selling at an average premium of 12% after adjusting for other attributes, reflecting the market's appreciation for heritage characteristics, architectural style, and official listing status [26]. In Texas, homes in historically designated areas were also found to sell at a premium [27]. Analyzing property transactions in Auckland, New Zealand, from 2006 to 2016, local heritage protections were associated with a 1.7% price increase within 50 m of heritage sites, 1.4% within 100 m, and 0.5% within 200 m, while properties in distinctive areas experienced a 4.3% increase [28]. Moreover, research in Chicago, Illinois, indicates that heritage preservation impacts vary by authority; national policies generally positively affect property values, while local policies may sometimes yield an adverse effect [29]. Supporting this, local preservation efforts were found to negatively impact property values in Philadelphia, Pennsylvania [30].

Research on human settlement heritage and the impacts of heritage designation on settlements in Vietnam is limited, with a few studies offering insights but underscoring gaps. Nixon [31] underscored the demand to address infrastructure gaps while preserving the residents' cultural ties to the site in Hue Citadel. Similarly, Quang [32] revealed that heritage conservation practices of the Cham temples in Ninh Thuan province do not include indigenous living heritage and tangible aspects of local culture. This gap indicates a need for broader research on balancing heritage preservation, community benefit, and sustainable development within Vietnam's heritage-rich areas.

In the context of the Trang An Landscape Complex—a World Heritage site—no prior research has comprehensively assessed how heritage status and 'place-based' factors influence human settlement value. To address this gap, the primary objective of this study is to evaluate the economic value of human settlement within the complex and to analyze the specific determinants that drive these values. To achieve this, the study

focuses on three key research questions: (1) What is the estimated human settlement value across the core, buffer, and surrounding zones of the heritage complex? (2) To what extent do intangible factors, such as resident perception and mobility, influence property values alongside traditional structural and locational attributes? and (3) How do these value determinants vary spatially across the different management zones? By answering these questions, this research aims to highlight the economic significance of cultural resources and provide actionable insights for integrating human settlement values into sustainable development strategies.

2 Methodology

2.1 Place-oriented human settlement value

Place-oriented human settlement value is defined as a holistic assessment framework that reconceptualizes settlements not merely as static physical structures or spatial locations, but as dynamic "places" infused with human meaning, social interaction, and lived experience [33, 34]. This approach comprises two additional core components: human dynamics, which quantify spatiotemporal mobility patterns (such as hourly visit frequencies) to reflect how environments are actively utilized, and human perceptions, which typically employ deep learning on street-view imagery to measure the subjective "sense of place" through dimensions like safety, liveliness, and beauty [3].

Figure 1 illustrates the theoretical framework comparing the Conventional Perspective (space-based) with the Place-based Perspective for evaluating human settlements. The Conventional Perspective relies on static and objective metrics—specifically Structural Attributes (e.g., area, baths, built year) and Locational Attributes (proximity to schools, stores, hospitals)—to estimate the value of the physical "House." In contrast, the Place-based Perspective integrates the intangible, human-centric dimensions of the "Home," capturing Human Dynamics (mobility patterns) and Human Perception (subjective qualities such as liveliness, safety, and beauty). The framework demonstrates that combining these physical "space" factors with experiential "place" factors allows for a more holistic evaluation of Human Settlement value, synthesizing the monetary value of real estate with the social and emotional value of the living environment [3].

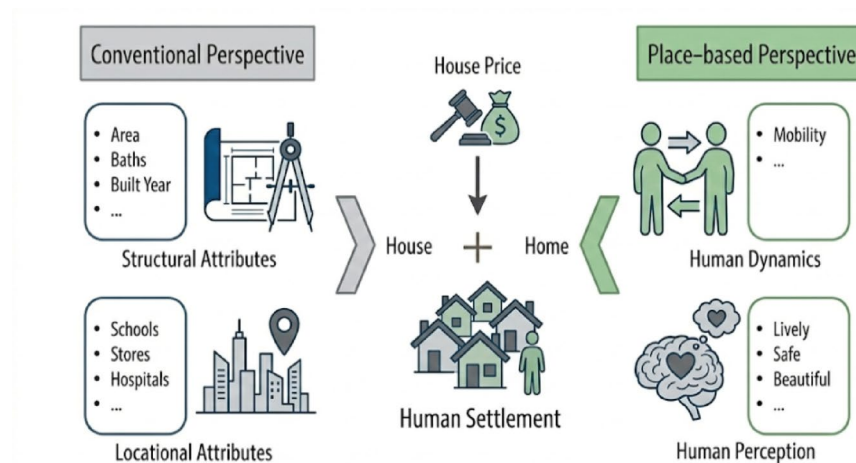


Fig. 1 Theoretical framework of place-oriented human settlement value.

Source: Adopted from Kang, et al. [3].

Figure 2 illustrates the study’s research framework, which integrates spatial econometric methods, including the Place-oriented hedonic price method (P-HPM) and Geographically weighted regression (GWR) with Monte Carlo simulation, to assess the factors influencing human settlement values in heritage areas. By incorporating P-HPM, this research explores how structural, location-based, and perceptual factors and mobility patterns influence housing prices. GWR is employed to account for spatial non-stationarity, capturing local variations in these relationships, while Monte Carlo simulation tests the robustness and spatial variability of the results. This framework enhances the understanding of how human behaviors, perceptions, and spatial factors shape housing prices, providing essential insights for urban planning, policy-making, and heritage conservation.

2.2 Placed-oriented hedonic pricing method (P-HPM)

Housing prices can serve as a representative of human settlement and economic conditions [3]. The Hedonic Price Method (HPM) is a valuation technique that analyzes diverse real estate market elements, such as rural and urban vacant land, single-family homes, and condominiums [35, 36]. Traditional HPM is derived from Lancaster [37] consumer theory and Rosen’s [38] theory. Both approaches seek to estimate the value of attributes by analyzing the relationship between the observed prices of products and their associated attributes.

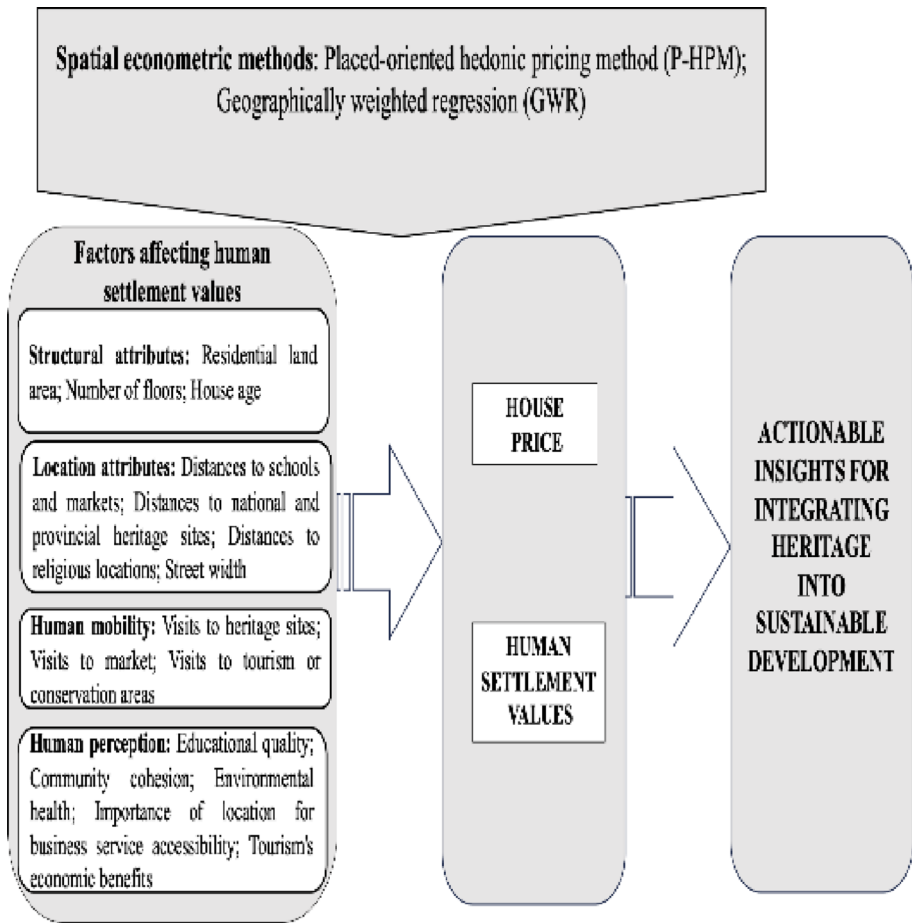


Fig. 2 Research framework

The application of the traditional HPM to evaluate the effects of heritage sites on real estate is widespread, as demonstrated by studies in Berlin, Germany [35], and the Netherlands [39]. Besides, some studies examined the impact of heritage (internal factors) and policies (external factors) on housing prices in historical conservation areas [26, 40]. Preserved historic real estate often exhibits unobserved characteristics, such as architectural and aesthetic features, which are associated with elevated property prices. The traditional HPM is commonly formulated as a multivariate regression model (Eq. 1):

$$P_H = \beta_0 + \beta_1 Struc + \beta_2 Loc + \varepsilon \quad (1)$$

where P_H represents the natural logarithm of the estimated housing prices, *Struc* denotes the structural attributes of the house, and *Loc* refers to the location attributes of the neighborhoods. Here, β_0 , β_1 , and β_2 are the corresponding coefficients estimated in the model, and ε represents the error term.

To investigate the spatial heterogeneity of impacts on real estate values, scholars have increasingly focused on studies examining the valuation of spatial amenities. A significant contribution to this field is the work of Lazrak, et al. [41] analyzing the economic effects of listed heritage sites on urban real estate values, drawing on a dataset of individual housing transactions spanning 22 years in Zaanstad, Netherlands. Similarly, Franco and Santos [42] examined the external impacts of different types of built heritage and their interactions with urban amenities in Lisbon, Portugal.

HPM models for residential property valuation are widely utilized and are fundamentally proposed from a spatial perspective, primarily focusing on static housing structural information and objective building environment factors. However, the value of real estate is not solely determined by its spatial context; it also varies from location to location, influenced by different cultural backgrounds, human motivations, perceptions, and social interactions [3]. Thus, this paper adopted a place-oriented hedonic pricing model (P-HPM) incorporating human motivations and perceptions to enhance understanding of settlement patterns in the traditional HPM (Eq. 2). This approach provides fresh insights into the value of human settlement by highlighting the significance of place, utilizing geographic data. The P-HPM is articulated as follows:

$$P_H = \beta_0 + \beta_1 Struc + \beta_2 Loc + \beta_3 Vis + \beta_4 Percept + \varepsilon \quad (2)$$

"*Vis*" represents the model of human visits to locations, while "*Percept*" refers to human perceptions of those locations.

From the P-HPM perspective, specific variables are integrated estimated models for land and residential properties within the study area. These groups include structural attributes, location attributes, human perception, and human mobility factors, each adding distinct insights into how different characteristics influence real estate values within this heritage setting.

Structural attributes describe physical property characteristics influencing intrinsic value. Residential land area (*S_Area*) directly affects prices, with larger plots generally valued higher. The number of floors (*S_Floor*) indicates vertical development, often correlating with increased value. House age (*S_Builtyear*) typically favors newer constructions, although older properties in heritage areas may gain value from historical or cultural significance.

Location attributes capture proximity (measured in meters) to essential amenities and heritage landmarks, impacting accessibility and cultural appeal. Distances to schools (L_School) and markets (L_Market) reflect convenience and accessibility valued by residents. Closeness to national ($L_Nationalheritage$) and provincial ($L_Provincialheritage$) heritage sites and religious locations (L_Temple) enhances cultural significance and property desirability. Street width ($L_Streetnarrow$) influences accessibility, with wider streets typically boosting property values.

Human perception variables (Likert scale, 1–5) represent residents' subjective views impacting property values. Perceptions of educational quality ($P_Education$) influence area attractiveness, especially for families. Community cohesion ($P_Neighbor$) indicates neighborhood appeal, while perceived environmental health ($P_Environment$) relates to living quality. Perceived importance of location along major thoroughfares for business service accessibility ($P_Service$). Views on tourism's economic benefits ($P_Tourism$) reflect growth expectations and area desirability.

Human mobility variables measure residents' interactions within cultural, consumer, and tourism spaces (hours/month). Time spent visiting heritage sites ($V_Heritage$) indicates cultural engagement; market visits (V_Market) represent routine consumer interactions, while visits to tourism or conservation areas ($V_Tourism$) reflect participation in tourism-related activities. These interactions dynamically influence property valuation within the culturally diverse landscape.

2.3 Geographically weighted regression (GWR) and Monte Carlo simulation

Conventional HPM approaches often utilize a multi-linear regression model with OLS estimation, assuming spatial stationarity—meaning that the relationships between house prices and influencing factors remain consistent across the area. However, these relationships can vary spatially, leading to significant variations in parameter estimates [43, 44]. This study adopted Geographically Weighted Regression (GWR) to capture these spatial variations in relationships, the typical GWR model form is presented in Eq. 3.

$$Y_i = \alpha_{0(u_i, v_i)} + \sum_{k=1}^m \alpha_{k(u_i, v_i)} X_{k(u_i, v_i)} + \varepsilon_i \quad (3)$$

In this model, Y_i represents the natural logarithm of the average house price at location i , with coordinates (u_i, v_i) ; $\alpha_{0(u_i, v_i)}$ is the intercept term; $\alpha_{k(u_i, v_i)}$ denotes the local regression coefficient for the k -th independent variable; $X_{k(u_i, v_i)}$ represents the k -th attribute at location i ; and ε_i is the random error term.

The primary goal of employing GWR in this study is to examine the presence of spatial non-stationarity—that is, to determine if the relationships between housing prices and place-related variables differ across locations. To verify the significance of spatial non-stationarity, a Monte Carlo test is typically recommended [3]. This test assesses whether the local estimators exhibit substantial spatial variation, allowing us to evaluate the reliability of the results.

2.4 Study area

The Trang An Landscape Complex (Fig. 3) comprises numerous heritage sites and scenic landscapes that have been classified across all levels of heritage designation: provincial, national, and special national levels. This includes two special national heritage

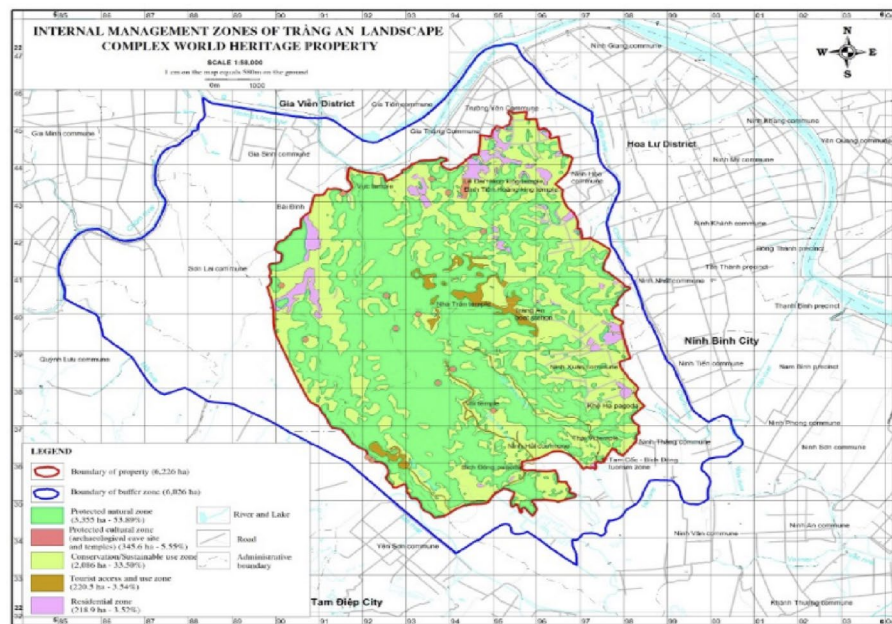


Fig. 3 Management zone of Trang An landscape complex [1]

sites—Hoa Lu Ancient Capital and Trang An – Tam Coc – Bich Dong—along with 21 national and 18 provincial-level heritage sites. The land within the heritage site is strictly protected under Vietnamese law and is directly managed by the Trang An Landscape Complex Management Board. These areas are safeguarded following regulations governing national and special national monuments, as outlined in the Law on Cultural Heritage [1, 45, 46].

The majority of the population in the core zone (approximately 44,000 residents) and buffer zone of Trang An Landscape Complex engages in agriculture and traditional crafts, particularly stone carving. Most inhabitants are dispersed outside the valleys and limestone mountain ranges of Hoa Lu. The local community plays a vital role in preserving the region's cultural traditions and identity, while also serving as an important human resource for cultural tourism. They not only supply labor for tourism activities but also contribute to developing local services. It is essential to integrate tourism activities with the local community, enabling them to participate in and benefit from these services, thereby fostering a sense of responsibility in protecting and developing this unique heritage area [1].

The Trang An Landscape Complex holds exceptional historical, cultural, military, scientific, educational, and ecological significance in Vietnam. Historically, it includes Hoa Lu, Vietnam's ancient capital, symbolizing national unity during the Dinh, Early Le, and Ly dynasties. Culturally, it features numerous temples, archaeological sites, and landmarks linked to Vietnamese legends (Fig. 4). Strategically designed for defense, Hoa Lu leveraged its natural terrain for protection and supported local agriculture and crafts. Scientifically, the complex showcases sophisticated architecture from various dynasties, valuable for historical research. Educationally, it fosters national pride and cultural exchange. Ecologically, Trang An boasts a rich biodiversity, with 618 plant species and endangered wildlife, such as the white-cheeked gibbon, underscoring its global ecological importance.

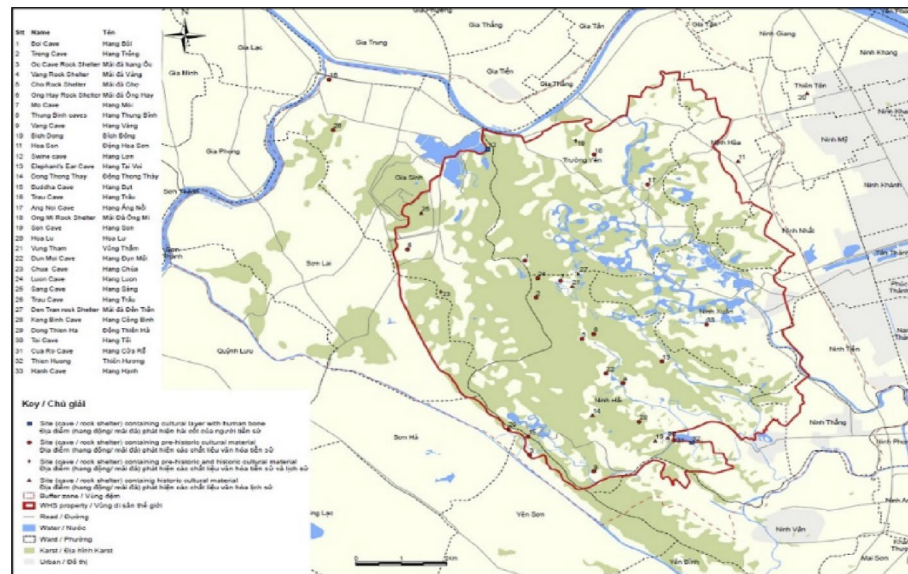


Fig. 4 Archaeological sites in the Trang An Landscape Complex [2]

2.5 Data collection

Qualitative and quantitative data were gathered through face-to-face interviews with key informants (KIs) and household interviews in August 2024. The study utilized a stratified random sampling method to determine the sample size for household interviews. Based on Yamane's [47] equations and the total of approximately 4000 households in three communes of the Trang An Landscape Complex area (VietNam Prime Minister, 2016)—a minimum of 300 households was deemed necessary to ensure a 95% confidence level. The formula employed is as follows:

$$n = \frac{N}{1 + N(e)^2} \quad (5)$$

where n represents the sample size, N denotes the population, and e indicates the reliability level.

To ensure a sufficient number of representative samples for the study, accounting for potential unreliability in some samples, a total of 320 survey samples will be collected in 3 communes of Hoa Lu district (Ninh Hai, Ninh Xuan, and Truong Yen) and two wards of Ninh Binh city (Tan Thanh and Ninh Nhat) (Fig. 5).

Aside from households in the core and buffer zone of the complex, houses in the surrounding zone were also surveyed. The surrounding zone samples offer insight into the influence of heritage designation on areas outside the formal boundaries of preservation but are still economically and socially connected to the heritage complex. This three-zone approach captures a range of heritage and development interactions, allowing for a more nuanced understanding of how proximity to heritage sites affects property values and settlement dynamics.

The criteria for sample selection included: (1) residential areas within the Tam Coc—Bich Dong tourist zone; (2) homestays designed in the style of traditional houses, complemented by aesthetically pleasing and modern garden landscapes; (3) households involved in agriculture and tourism activities, such as boat rowing; (4) households located on main roads that provide convenient access for business services; (5)

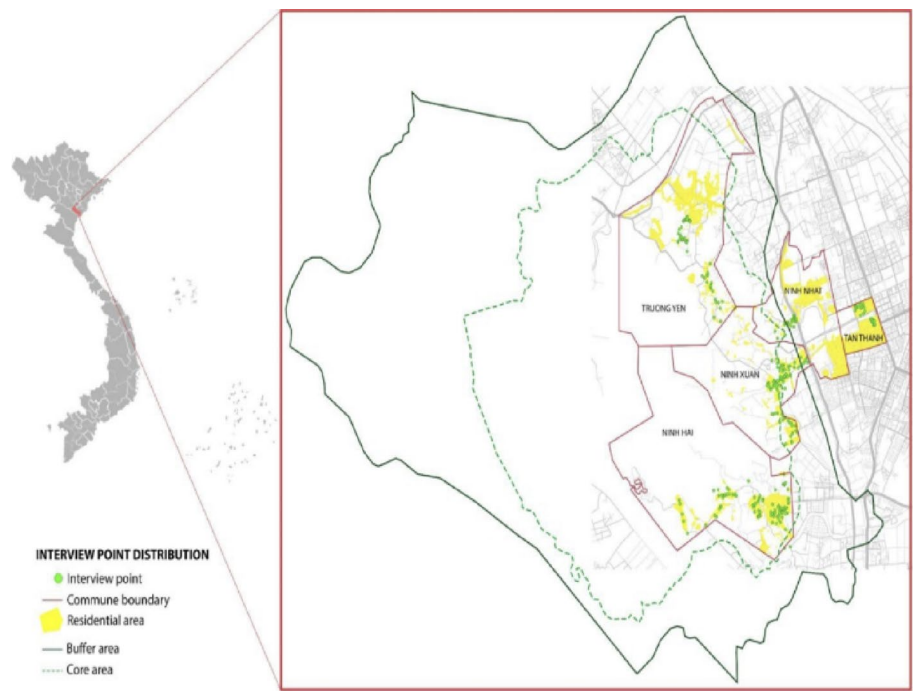


Fig. 5 Study sites and interview point distribution.
Source(s): Authors’ work, 2024

Table 1 Number of participants in the household survey and KI interviews.
Source(s): Authors’ own work, 2024

Location	Household survey	KIs
Ninh Hai commune, Hoa Lu district	101	Ninh Binh Department of Tourism (1), Ninh Binh DONRE (1)
Ninh Xuan commune, Hoa Lu district	91	Trang An Landscape Complex Management Board (1)
Ninh Nhat ward, Ninh Binh city	25	Urban Management Division of Ninh Binh City (1), Natural Resources and Environment Division of Hoa Lu District (1)
Truong Yen commune, Hoa Lu district	60	Economic and Infrastructure Division of Hoa Lu District (1), Gia Vien District (1), and Nho Quan District (1)
Tan Thanh ward, Ninh Binh city	30	Land Management Officers in Core and Buffer Zone Communes (17)
Total	307	25

households situated near significant temples, communal houses, or cultural centers; (6) households with scenic landscapes (e.g., rivers, lakes, rice fields, and hills); (7) households engaged in traditional embroidery; (8) historical houses; and (9) families lacking the aforementioned characteristics.

Additionally, survey samples were distributed across communes and wards to ensure representativeness and reliability in the estimation results (Table 1). This process involved selecting households based on various criteria, including road frontage, accessibility, proximity to tourist attractions, and suitability for business operations. To facilitate the creation of a land price distribution map, the sampling process incorporated the determination of geographical coordinates at the survey site using Timestamp Camera software, requiring the recording of coordinates at the yard of the surveyed households. If a yard was unavailable, coordinates were taken at the center of the house.

The household survey questionnaire was developed to gather the following information: Demographic and socioeconomic characteristics (age, gender, occupation, education, income); Property characteristics and rights; Determinants of fluctuations in property prices; Property attributes; Locational attributes of property; Community perceptions of location; Frequency of visits to destinations. In each commune and ward, the household survey questionnaire was subjected to a preliminary direct survey involving eight randomly selected household heads, adhering to established sampling criteria. This process enabled the research team to refine specific questions in the questionnaire, thereby enhancing its relevance and overall quality. Subsequently, a survey was conducted with 320 households using the revised questionnaire. The collected responses were rigorously reviewed, excluding those with incomplete or unreliable information. After removing unreliable responses (incomplete or inaccurate data), 307 valid survey responses were retained for the analysis, as detailed in Table 1.

KI interviews were also carried out with people closely involved in the management and strategic decision-making process for the sustainable development of Trang An Landscape Complex. This helps the research gain deeper insights into the vision, opportunities and challenges of effectively managing the region. Interviewing 25 KIs provides the research with a deeper foundation for determining the actual market prices of residential land and housing (refer to Table 1).

3 Results

3.1 Demographic characteristic of the sample

The local respondents ranged in age from 20 to 93 years, with an average age of 54, aligning well with the study's target demographic of adult residents in the area. The gender distribution was relatively even, with 166 male respondents (54.07%) and 141 female respondents (45.93%). This balanced gender representation supports the inclusion of both and female perspectives in the analysis of human perceptions and mobility within the heritage complex. Educational attainment varied, with the most significant proportion completing secondary school (128 individuals, 41.69%), followed by those who finished high school (91 individuals, 29.64%). Additionally, 26 participants held a university degree (8.47%), while a small portion (0.65%) had postgraduate qualifications (Table 2). These varying educational levels are crucial for understanding how education influences

Table 2 Descriptive statistics of household characteristics.
Source(s): Survey data processing results, 2024

Value		N	Frequency	Percentage
Gender	Male	307	166	54.07
	Female	307	141	45.93
Marital status of household head	Married	307	301	98.05
	Single	307	6	1.95
Educational level	No formal education	307	15	4.89
	Primary education	307	19	6.19
	Lower secondary education	307	128	41.69
	Upper secondary education	307	91	29.64
	Vocational training	307	12	3.91
	College	307	14	4.56
	University	307	26	8.47
	Postgraduate	307	2	0.65

respondents' perceptions of the heritage site and their ability to engage with complex mobility and development issues.

Respondents' occupation types were diverse, reflecting the local economy's reliance on a combination of agriculture and tourism (Table 3). Most earned income from service-oriented activities (50.16%), such as running restaurants, operating homestays, working as hotel staff, boat rowers, or shopkeepers. These occupations are particularly relevant to the study as they are closely connected to mobility and perceptions of heritage sites, with tourism workers frequently interacting with visitors and engaging with the heritage area.

Agriculture was the second-most common income source, with 38.76% of households engaged in farming. Furthermore, 22.48% of households reported having at least two income sources, typically agriculture and tourism-related services, as locals often farm rice or engage in other agricultural activities while participating in tourism during off-peak seasons. This dual-income structure illustrates the interdependence between different economic sectors in the region, and how such occupational patterns influence both mobility and residents' engagement with the heritage site.

Beyond these primary sources, 6.51% of households received additional income through local allowances, support for elderly family members, or pensions for veterans. On average, households earned approximately 21.04 million VND monthly from tourism activities. Household incomes ranged widely, from those with no formal income (supported by family) to some earning up to 300 million VND monthly, highlighting a significant income disparity within the region. This income disparity highlights the socioeconomic diversity within the sample, suggesting that variations in income could play a significant role in shaping both mobility patterns and perceptions of heritage preservation.

Respondents' involvement in local tourism activities further illuminates the relationship between mobility and heritage perceptions. A total of 66.78% of households

Table 3 Income sources of the surveyed households.

Source(s): Survey data processing results, 2024

Value		N	Frequency	Percentage
Occupation	Agriculture	307	119	38.76
	Government official	307	37	12.05
	Small business	307	34	11.07
	Construction	307	10	3.26
	Fisherman	307	2	0.65
	Services	307	154	50.16
	Factory worker	307	5	1.63
	Traditional crafts	307	5	1.63
	Other	307	20	6.51
Participation in local tourism activities	Yes	307	205	66.78
	No	307	102	33.22
Tourism activities the household participated in	Participated in planning, regulation of local tourism activities	205	11	5.37
	Participated in cultural, art, and festival activities	205	16	7.80
	Transportation services	205	62	30.24
	Small business	205	42	20.49
	Organized production, traditional crafts demonstration	205	2	0.98
	Tour guide	205	4	1.95
	Contributed capital, resources	205	90	43.90
	Other	205	1	0.49

participated in some form of tourism-related activity. The types of involvement varied, with many respondents contributing to tourism planning or participating in cultural, art, and festival activities. The most common form of participation was through transportation services (30.24%), followed by small businesses (20.49%) and capital contribution or resource sharing (43.90%). These activities underscore the active participation of local population in tourism, reinforcing the relationship between economic activities, mobility, and heritage site development.

The household survey reveals significant zone-based differences (Table 4). Property area is largest in the core (215.94 m²), followed by buffer (207.19 m²) and surrounding zones (121.33 m²) ($p < 0.001$). Surrounding zone properties have more floors (2.67) compared to core (1.61) and buffer zones (1.66) ($p < 0.001$). The surrounding zone has closest proximity to schools and markets, while buffer zone is farthest from schools ($p < 0.001$). Perceptions of education, neighbor relations, and environment are positive overall. Heritage visitation frequency is highest in the surrounding zone (11.59 h/month; $p = 0.004$), whereas tourism site usage peaks in the buffer zone (21.23 h/month; $p < 0.001$).

3.2 P-HPM model estimation

The estimation results of the P-HPM model (Table 5) reveal distinct effects of various factors on real estate prices within each zone. In the core zone, the area variable has a strong positive and significant effect ($\beta = 0.404$, $p < 0.01$), indicating that larger areas are associated with higher prices. The influence of proximity to heritage sites is

Table 4 Characteristics of houses and respondents.

Source(s): Authors' work, 2024

Characteristic	Overall N = 307 ¹	Core zone N = 191 ¹	Buffer zone N = 86 ¹	Surrounding zone N = 30 ¹	p-value ²
Housing price (Mil. VND/m ²)	40.765 (22.450)	33.272 (16.093)	49.372 (27.884)	63.800 (14.291)	< 0.001
Land area (m ²)	204.241 (93.847)	215.937 (91.089)	207.186 (93.347)	121.333 (70.658)	< 0.001
Number of floor	1.725 (0.891)	1.605 (0.869)	1.663 (0.791)	2.667 (0.758)	< 0.001
Built year	29.450 (162.498)	37.545 (205.624)	16.267 (11.660)	15.700 (8.289)	0.6
Street narrow (m)	5.352 (3.691)	4.861 (2.998)	4.879 (2.621)	9.817 (6.456)	< 0.001
School distance (m)	5.812 (2.881)	5.672 (1.809)	7.845 (3.121)	0.882 (0.268)	< 0.001
Market distance (m)	0.935 (1.068)	1.095 (1.202)	0.836 (0.787)	0.200 (0.090)	< 0.001
Provincial heritage distance (m)	1.287 (0.877)	1.290 (0.877)	1.243 (0.947)	1.388 (0.663)	0.2
National heritage distance (m)	1.300 (0.635)	1.278 (0.656)	1.285 (0.618)	1.487 (0.530)	0.2
Temple distance (m)	0.842 (0.617)	0.830 (0.603)	0.772 (0.566)	1.125 (0.772)	0.057
Heritage frequency (hours/ month)	10.255 (22.789)	10.046 (24.772)	10.256 (15.978)	11.590 (26.631)	0.004
Market frequency (hours/month)	25.822 (23.282)	26.478 (26.256)	24.445 (18.110)	25.597 (15.282)	0.3
Tourism places frequency (hours/ month)	10.483 (27.930)	5.181 (10.694)	21.225 (48.075)	13.444 (13.094)	< 0.001
Education perception	3.733 (0.605)	3.696 (0.617)	3.721 (0.546)	4.000 (0.643)	0.066
Neighbor perception	4.124 (0.619)	4.110 (0.652)	4.128 (0.570)	4.200 (0.551)	0.8
Service perception	3.941 (0.826)	3.979 (0.781)	3.930 (0.865)	3.733 (0.980)	0.6
Tourism perception	4.163 (0.828)	4.199 (0.816)	4.186 (0.759)	3.867 (1.042)	0.3
Environment perception	4.342 (0.669)	4.309 (0.714)	4.360 (0.612)	4.500 (0.509)	0.5

¹Mean (SD)

²Kruskal-Wallis rank sum test

Table 5 Estimation of the P-HPM model.

Source(s): Authors' own work, 2024

	Core zone	Buffer zone	Surrounding zone
S_Area	0.404*** (0.061)	0.214* (0.119)	0.439*** (0.085)
S_Floor	0.079 (0.058)	−0.001 (0.124)	−0.246** (0.095)
S_Builtyear	−0.013 (0.023)	0.021 (0.053)	0.081 (0.046)
L_Streetnarrow	−0.013 (0.049)	0.306*** (0.101)	0.250*** (0.071)
L_School	0.058 (0.060)	0.101 (0.056)	−0.016 (0.149)
L_Market	−0.023 (0.026)	0.096* (0.052)	0.324*** (0.059)
L_Provincialheritage	−0.050 (0.036)	−0.173*** (0.056)	−0.041 (0.063)
L_Nationalheritage	−0.119*** (0.036)	−0.235*** (0.087)	−0.176** (0.059)
L_Temple	0.093*** (0.030)	0.124** (0.051)	0.075* (0.035)
V_Heritage	0.052*** (0.015)	0.081*** (0.029)	−0.006 (0.016)
V_Market	0.010 (0.022)	0.047 (0.037)	−0.085 (0.057)
V_Tourism	0.028* (0.015)	0.025 (0.021)	0.005 (0.021)
P_Education	0.323** (0.134)	0.450 (0.336)	0.106 (0.229)
P_Neighbor	0.219 (0.148)	0.702** (0.339)	0.615** (0.216)
P_Service	0.111 (0.122)	0.167 (0.179)	0.656*** (0.109)
P_Tourism	0.241** (0.120)	0.293 (0.213)	0.346** (0.119)
P_Environment	0.260* (0.139)	0.420 (0.317)	−0.389 (0.281)
Constant	13.190*** (0.456)	12.419*** (1.054)	14.786*** (0.723)
Observations	191	86	30
R ²	0.540	0.700	0.972
Adjusted R ²	0.494	0.625	0.933
Residual Std. Error	0.317 (df=173)	0.358 (df=68)	0.112 (df=12)
F Statistic	11.931*** (df=17; 173)	9.348*** (df=17; 68)	24.841*** (df=17; 12)

***, **, * Significant at the 1%, 5%, and 10%, respectively

reflected in the negative and significant impact of the distance to national heritage sites ($\beta = -0.119$, $p < 0.01$). Additionally, the perception of tourism and environmental perception positively impact property prices, suggesting that residents value preserving cultural and environmental quality within the core area. Also, the visiting times to heritages ($\beta = 0.052$) and tourist places ($\beta = 0.028$) significantly affect house prices at 5% and 10%, respectively.

In the buffer zone, several accessibility factors significantly affect house prices. Notably, narrow streets ($\beta = 0.306$, $p < 0.01$) and proximity to markets ($\beta = 0.096$, $p < 0.10$) are positively associated with prices. The negative influence of distance from provincial and

national heritage sites highlights the value residents place on heritage accessibility in this area. Neighbor perception ($\beta = 0.702$, $p < 0.05$) also significantly impacts prices, indicating that social cohesion is essential to the buffer zone. Also, the visiting time to heritages ($\beta = 0.081$, $p < 0.05$) positively influences house prices in this zone.

In the surrounding zone, property area remains a significant determinant of house prices ($\beta = 0.439$, $p < 0.01$). Access to markets has a strong positive influence ($\beta = 0.324$, $p < 0.01$), while distances to heritage sites (both national and provincial) negatively impact property values, although at different significance levels. The presence of narrow streets also shows a positive impact on prices in this area. Furthermore, perception variables, such as neighbor perception ($\beta = 0.615$, $p < 0.05$), service perception ($\beta = 0.656$, $p < 0.01$), and tourism perception ($\beta = 0.346$, $p < 0.05$), indicate that community-based and service-oriented perceptions significantly affect property values in the surrounding zone.

The estimated function representing the relationship between human settlement value and settlement area in the Trang An landscape complex can be summarized using the average values of variables in Table 4 as follows:

$$\text{Corezone : Price} = e^{13.190} \cdot (S_Area)^{0.404} \cdot 3.518948$$

$$\text{Bufferzone : Price} = e^{12.419} \cdot (S_Area)^{0.214} \cdot 5.495773$$

The total human settlement value of the Trang An landscape complex is derived by summing the settlement values in the core and buffer zones. The settlement areas of the core and buffer zones are 349 hectares and 896 hectares, respectively¹. The human settlement value for each zone was calculated as follows:

$$\begin{aligned} \text{Corezone : } & \int_{70}^{3490000} e^{13.190} \cdot S_Area^{0.404} \cdot 3.518948 \cdot d(S_Area) \\ & = 2,058,338 \text{Billion VND} \sim 84.39 \text{Billion USD} \end{aligned}$$

$$\text{Bufferzone : } \int_{80}^{8960000} e^{12.419} \cdot S_Area^{0.214} \cdot 5.495773 \cdot d(S_Area) = 308.613.4 \text{Billion VND} \sim 12.65 \text{Billion USD}$$

Thus, the total human settlement value of the Trang An landscape complex is approximately 95.04 billion USD.

3.3 Geographically weighted regression

We assessed the spatial variability of the estimated coefficients through the GWR model and a Monte Carlo significance test. Table 6 reports the p -values for each variable in the core, buffer, and surrounding zones, derived from 99 spatial permutations and one actual spatial arrangement. Most variables exhibited non-significant p -values in all zones, indicating no significant spatial variability. This suggests consistent impacts on property values across spatial contexts.

In contrast, two location-related variables display significant spatial variability across zones. Specifically, distance to the market and distance to the temple in the core zone show p -values below 0.05, suggesting that these locational attributes' influences on housing values vary largely from place to place, and it is necessary to model the spatial

¹ 1 \$ = 24,390.24 VND.

Table 6 GWR-based place-oriented hedonic pricing model with Monte Carlo significance test.
Source(s): Authors' work, 2024

Variables	Core zone	Buffer zone	Surrounding zone
	<i>p</i> -value	<i>p</i> -value	<i>p</i> -value
(Intercept)	0.17	0.04	0.45
S_Area	0.18	0.37	0.48
S_Floor	0.17	0.91	0.54
S_Builtyear	0.94	0.36	0.14
L_Streetnarrow	0.27	0.05	0.90
L_School	0.26	0.99	0.26
L_Market	0.00	0.37	0.61
L_Provincialheritage	0.98	0.2	0.45
L_Nationalheritage	0.46	0.81	0.89
L_Temple	0.03	0.36	0.66
V_Heritage	0.27	0.02	0.86
V_Market	0.18	0.04	0.63
V_Tourism	0.57	0.06	0.33
P_Education	0.14	0.6	0.43
P_Neighbor	0.96	0.49	0.06
P_Service	0.95	0.58	0.16
P_Tourism	0.70	0.47	0.54
P_Environment	0.98	0.58	0.64

heterogeneity of these variables. The distance to amusement parks was also found to have a spatially heterogeneous impact in the study of Kang, et al. [3]. Additionally, heritage visitation frequency in the buffer zone also exhibits a significant *p*-value ($p = 0.02$), indicating spatial variation in how cultural engagement affects property values.

4 Discussion

4.1 The interconnection with heritage-related variables significantly affects the human settlement value

The results of the P-HPM model underscore the intertwined relationship between cultural heritage, resident experiences, and property market dynamics, providing insights into how heritage and social factors shape real estate values in this unique landscape.

First, proximity to heritage sites consistently emerges as a key determinant of property values. In both the core and buffer zones, properties located closer to national and provincial heritage sites command higher prices, reflecting the cultural and aesthetic appeal associated with these areas. This finding suggests that residents and potential buyers value living near historically significant and scenic locations, underscoring the premium placed on properties within or near heritage landscapes. In the core zone, specifically, the negative coefficient for the distance to national heritage sites highlights that closer proximity to these landmarks increases real estate value, demonstrating a strong attachment to heritage elements. In the Dutch city of Zaanstad, buyers are willing to pay a 26.9% premium for acquiring a monument, while nearby properties see a 0.28% increase in value for each additional monument within a 50-m radius. Furthermore, properties within historically protected areas enjoy a 26.4% premium, highlighting the significant impact of a historic ensemble effect [48]. In Auckland, New Zealand, properties located within special character areas command a 4.3% price premium [28]. This premium rises to 7.91% for properties within both a heritage area and a special character area,

indicating a heightened value associated with these combined designations. Other studies also confirmed the positive externalities of heritage areas on property prices [27, 49, 50].

Perception variables also play a notable role, albeit with different impacts across zones. In the core zone, positive perceptions of educational quality, neighbor relations, and tourism contribute positively to property values, reflecting the influence of community quality and cultural appeal. This suggests that residents perceive living in the core zone as an opportunity for enhanced quality of life, closely tied to the heritage environment and community connections. In the buffer and surrounding zones, perceptions regarding neighbors and services are significant, indicating that social cohesion and access to essential services are particularly valued in areas further from the core heritage sites. The perception of tourism's positive impact in the core and surrounding zones also indicates that residents recognize tourism as an economic asset that enhances property values, especially in areas with cultural appeal. This finding demonstrates the deep connection between human settlement and 'sense of place,' as this sense extends beyond human mobility and perception to encompass social experiences, emotions, place-based understanding, and cultural constructs [51, 52]. Additionally, frequency of visits to heritage and tourism sites has a noticeable impact on property values, particularly in the core and buffer zones. In the core and buffer zones, frequent engagement with heritage sites positively influences property values, suggesting that regular interaction with cultural landmarks enhances residents' sense of place and the value they assign to properties in proximity to these sites. In core zone, tourist places visiting time shows a positive association with property values, implying that areas closer to tourist attractions see added value from tourism-driven demand. This aligns with the fact that many properties in the buffer zone are used for tourism-related activities, catering to the influx of visitors seeking access to the scenic and cultural experiences that Trang An offers.

From a theoretical and managerial perspective, these findings suggest a critical pathway for future heritage governance: shifting from a purely 'monument-centric' conservation model to a 'place-oriented' value framework. Since human perceptions and mobility are statistically significant drivers of settlement value, preserving the physical landscape alone is insufficient. Effective management must actively cultivate the 'intangible' assets of the community—specifically social cohesion and environmental quality. Consequently, policy interventions should not only focus on physical restoration but also on 'soft' infrastructure investments (e.g., community centers, educational programs) that enhance resident perceptions. By elevating these perceptual values, authorities can intrinsically boost the economic resilience of the settlement without relying solely on external tourism revenue.

4.2 Challenges for the development and management of Trang An landscape complex regarding human settlement

The development and management of the Trang An Landscape Complex face several significant challenges, with multiple factors exerting pressure on this heritage site. From a policy perspective, this friction is manifested primarily through industrial pressures and urbanization, which challenge the integrity of the buffer zone as defined by UNESCO guidelines [53]. An overview of the policy framework and regulatory instruments governing the Trang An Landscape Complex is presented in Appendix 1.

Firstly, development pressures, particularly from limestone mining, pose a substantial risk to the environmental integrity of the complex. Currently, limestone extraction for cement production is conducted southeast of the heritage site, outside the buffer zone, with licensed extraction reaching approximately three million tons annually over a 19.5-hectare area. Additionally, five private operators can mine smaller quantities for construction and local handicrafts on 20 hectares within the buffer zone. The proximity of mining sites to the heritage area necessitates ongoing monitoring and strict compliance with environmental regulations because they may create transboundary impacts [54–56]. The province has ongoing negotiations to limit mining operations in the buffer zone, with long-term plans to phase out these activities and relocate cement factories. Such measures underscore the critical balance between economic interests and the need to safeguard the heritage landscape from industrial encroachment.

Agricultural practices also impact the heritage site, although on a smaller scale. Grazing activities within the site, particularly by goats and cattle, can potentially damage archaeological sites and the natural landscape. Awareness-raising initiatives are essential to mitigate these effects. Additionally, the seasonal burning of rice stubble, while temporary, contributes to air pollution, and the increasing use of chemical fertilizers and pesticides raises concerns about soil and water contamination. From an academic perspective, this represents a conflict between "living heritage" livelihoods and environmental standards [57, 58]. The issue of uncontrolled grazing further indicates a need for community-based management strategies that align local agrarian habits with archaeological site protection.

Natural water systems face threats from dredging-related erosion, sedimentation, illegal logging, and waste disposal. Sustainable dredging methods and coordinated waste management are vital for maintaining hydrological and ecological integrity. Promoting alternative energy sources could reduce reliance on local firewood, preserving forest cover.

Population growth, urbanization, and infrastructure expansion also pressure the heritage complex. Rapid urban development in nearby Ninh Binh City has increased demand for land for hotels, businesses, and other infrastructure. Although these developments are confined to areas outside the heritage site, unregulated expansion could threaten the landscape's aesthetic and cultural value. Approximately 14,000 residents live within the heritage site, primarily around the Hoa Lu Ancient Capital and several villages. Development within these areas is regulated to balance residents' needs with the heritage site's preservation objectives. The long-term management strategy encourages natural population growth, restricts new immigration, and promotes resettlement outside the heritage boundaries. These strategies aim to limit urban sprawl and ensure infrastructure development aligns with conservation goals, thus minimizing potential adverse impacts on the heritage landscape and ecosystem carrying capacity [59].

Finally, managing private accommodations, such as eco-resorts and homestays, involves strict oversight. Existing accommodations are regulated to ensure minimal environmental impact, with only small-scale eco-resorts operating under strict oversight. New developments or renovations undergo rigorous impact assessments and must be approved by management authorities, ensuring minimal environmental impact while providing essential tourism services.

Addressing these multi-faceted challenges requires a transition from reactive restrictions to proactive 'livelihood adaptation' strategies. To resolve the conflict between industrial mining and heritage integrity, the province should accelerate the implementation of a 'Green Transition Roadmap' for the buffer zone, offering vocational retraining and financial incentives for households shifting from extractive industries (cement/mining) to eco-tourism or high-value organic agriculture. Regarding agricultural pressures, a community-based 'payment for ecosystem services' (PES) model could be piloted, where farmers are subsidized to adopt traditional, chemical-free farming methods that align with the visual standards of the heritage site. Furthermore, to curb the risks of urbanization and infrastructure sprawl, a rigorous social impact assessment mechanism must be mandated for all new projects. Unlike standard environmental assessments, this assessment would specifically evaluate potential disruptions to the 'living heritage' fabric—ensuring that physical development does not come at the cost of the socio-cultural integrity that defines the Trang An community.

4.3 The dynamic changes in human settlement values due to future development plans

According to the "Comprehensive Report on the Ninh Binh Provincial Plan for the 2021–2030 Period, with Vision to 2050" [60], the objective is to sustainably manage, protect, and promote the site's exceptional cultural and natural values in compliance with World Heritage standards. This plan emphasizes the need to balance heritage conservation with nature protection, environmental sustainability, disaster preparedness, and economic growth, particularly through sustainable tourism (e.g., cultural tourism economy based on Indigenous knowledge, agricultural eco-tourism, etc.). It aims to address contemporary challenges related to the safeguarding of cultural heritage, management of human activities, and the development of essential infrastructure for both visitors and residents. The implementation of this plan is anticipated to result in significant changes in the values of human settlement. A major challenge in heritage management lies in balancing the economic goals of heritage sites with the need to ensure that local communities benefit from development, all while preserving the diverse physical and socio-cultural characteristics of the area [61]. Different goals among stakeholders will lead to increasing conflicts in heritage management conservation [62].

Development plans that expand infrastructure, such as roads, public services, and amenities, may increase the accessibility and attractiveness of properties within the buffer and surrounding zones. Improved connectivity could reduce travel time to heritage sites and urban centers, making these areas more desirable for both residential and tourism purposes. Enhanced infrastructure could drive property values up as accessibility improves, particularly in zones currently characterized by limited access. While this supports the policy goal of economic growth, it introduces the risk of "fragmentation," where linear infrastructure disrupts the ecological continuity of the core zone [63].

With plans for increased tourism activities, the surrounding and buffer zones may experience an economic boost, attracting investments in hospitality, retail, and tourism-related services. Higher visitor numbers could stimulate demand for commercial properties and encourage further residential development to accommodate workers in the tourism industry. This aligns with global trends where World Heritage designation acts as a brand, inflating local property values [64]. However, an influx of tourism could pressure local resources and infrastructure, potentially leading to environmental degradation

if not managed sustainably. The monitoring and managing of visitor flow was also found to damage the UNESCO World heritage landscapes in many countries [65, 66]. In the core zone, strict tourism management policies may need to be enforced to maintain the area's exclusive heritage appeal, balancing visitor numbers with conservation efforts to avoid diluting its cultural and economic value.

Population growth due to migration and urban expansion may increase demand for housing, particularly in buffer and surrounding zones where residential expansion faces fewer restrictions. As more people are drawn to areas near Trang An's cultural and natural assets, property values in these zones may rise, especially if residential developments are integrated with sustainable practices. In contrast, strict regulations in the core zone may restrict population growth, maintaining its premium value due to exclusivity and limited housing supply.

The expansion of heritage designations or enhanced preservation efforts could further elevate the cultural value of the complex, making properties near these sites increasingly desirable. If development plans emphasize the conservation of cultural landscapes, heritage appreciation could drive long-term stability and growth in human settlement values, especially within the core and buffer zones. Additionally, incorporating community involvement in conservation efforts could enhance residents' sense of ownership and responsibility, potentially leading to higher demand for properties with cultural significance. This will foster the development of a localized urban system that strengthens residents' ties to their land and community, promoting their autonomy while simultaneously preserving and enhancing the region's heritage and cultural values.

To navigate these dynamic changes and mitigate the risks of gentrification and fragmentation, the implementation of the 2021–2030 Provincial Plan must be anchored in an 'Inclusive Value Capture' framework. As infrastructure improvements and heritage appreciation drive up property values in the buffer and surrounding zones, there is a distinct opportunity to implement value-capture mechanisms (such as developer impact fees or a special heritage preservation tax on high-value commercial transfers). These revenues should be reinvested directly into the core zone communities who face the strictest development constraints, thereby balancing the economic inequities between zones. Additionally, to prevent the displacement of original residents due to rising costs—a phenomenon observed in other heritage sites—planning policies should introduce 'adaptive zoning' that caps the percentage of residential properties convertible to pure commercial tourist accommodation. This ensures that the 'living' nature of the settlement remains intact, allowing the local community to coexist sustainably with the burgeoning tourism economy rather than being displaced by it.

While this study establishes a robust model P-HPM for the Trang An Landscape Complex, several avenues for future research emerge to expand upon these findings. First, to enhance the precision of the mobility variables, future research could integrate big data sources, such as mobile GPS signaling or social media check-in data. This would provide granular, objective metrics on how residents and tourists interact with the landscape, complementing self-reported survey data. Besides, comparative research applying this framework to other UNESCO World Heritage sites in Vietnam (e.g., Hue Imperial City or Hoi An Ancient Town) and Southeast Asia would help establish a generalized theoretical model. Comparing these results would clarify how different heritage

typologies—cultural versus natural—differentially influence human settlement values, ultimately informing broader regional conservation policies.

5 Conclusion

This study provides a comprehensive analysis of human settlement values within the Trang An Landscape Complex, focusing on the intricate interplay between heritage designation and property valuation. The analysis estimates the total human settlement value at approximately 95.04 billion USD, with the core zone contributing 84.39 billion USD and the buffer zone 12.65 billion USD. Findings indicate that heritage designation enhances property values across core, buffer, and surrounding zones, with the strongest premiums observed in proximity to protected cultural and natural sites. The P-HPM reveals how variables such as proximity to heritage sites, tourism engagement, and environmental perceptions impact real estate values, underscoring the economic importance of heritage preservation.

Beyond physical and structural attributes, the P-HPM analysis reveals that settlement value is significantly influenced by "intangible" human factors. Residents' perceptions of environmental quality, community cohesion, and tourism benefits were found to be statistically significant drivers of property value. Furthermore, the frequency of residents' interactions with the landscape positively correlates with property value. This highlights that active cultural engagement and a strong "sense of place" actively enhance the economic worth of the settlement, proving that value is derived not just from the physical existence of monuments, but from the community's lived experience within them.

The UNESCO World Heritage designation of the Trang An Landscape Complex is a national pride for the people of Vietnam and Ninh Binh province, who share the responsibility of safeguarding its globally significant values. This prestigious status calls for a sustainable development approach that balances heritage conservation with community engagement and economic resilience.

Development policies in Ninh Binh should prioritize enhancing human settlement values that align with heritage conservation. By positioning human settlement values at the heart of development strategies, policies can reinforce the economic and cultural importance of preserving heritage sites, ultimately supporting sustainable growth across the province.

Local communities, especially those residing in core and buffer zones, should be made fully aware of the heritage values they are entrusted to protect. As the primary stewards of the heritage site, local residents have a crucial role in choosing to protect and preserve these sites instead of opting for alternative land uses or sales. Raising awareness of the significance of heritage can foster a deeper community commitment to conservation efforts. Since residents in core and buffer zones face more stringent development restrictions than those in the surrounding zone, equitable development strategies must be implemented. These strategies should carefully consider the development rights of communities in each zone, ensuring that restrictions align with conservation goals and offering economic benefits opportunities. Sustainable development approaches that respect the balance between conservation and community development can help mitigate potential inequalities between zones.

A proactive, collaborative approach involving government agencies, research institutions, businesses, and local communities is essential for safeguarding Trang An's cultural

and natural heritage for future generations. Recognizing and respecting the rights, privileges, and traditional livelihoods of local residents encourages their active participation in heritage management, thereby enhancing residents' quality of life and fostering greater stewardship of the landscape.

Despite the robust findings, this study is subject to several limitations. First, the research employs a cross-sectional design based on data collected in August 2024, which captures property values and resident perceptions at a single point in time. Consequently, the analysis cannot identify causal relationships or track long-term trends in value fluctuation or displacement, limiting the ability to observe how these values evolve in response to ongoing policy changes. Second, the reliance on self-reported data for human mobility and perception variables introduces potential subjectivity and recall bias, as residents' reported behaviors may not always perfectly align with their actual daily patterns. Finally, the model focuses primarily on micro-level determinants (structural, locational, and perceptual), potentially overlooking broader macro-economic factors—such as regional or national real estate policies—that also exert influence on human settlement values.

Appendix 1. Overview of the policy framework and regulatory instruments governing the Trang An landscape complex

No	Document No	Date	Issuing Authority	Content Summary
1	82/QĐ-TTg	29/04/2003	Prime Minister	Approving the Master Plan for conservation, restoration, and promotion of values of the Hoa Lu Ancient Capital historical-cultural relic area
2	1875/QĐ-UB	10/08/2004	Ninh Binh PPC	Approving the Detailed Planning of Trang An Tourism Area
3	08/2005/NĐ-CP	2005*	Government	Government Decree on Construction Planning
4	2570/QĐ-UB	18/11/2005	Ninh Binh PPC	Approving the Planning of Trang An Tourism Area (Adjusted & Supplemented)
5	746/QĐ-UBND	12/04/2006	Ninh Binh PPC	Approving the Master Plan of Ninh Binh urban system and rural residential areas until 2020
6	230/UBND-VP5	18/07/2007	Ninh Binh PPC	Official Letter to the Ministry of Construction regarding the planning of Trang An Tourism Area
7	443/QĐ-BXD	31/03/2008	Ministry of Construction	Approving the tasks for the General Construction Planning project of Trang An Tourism Area
8	105/2008/QĐ-TTg	21/07/2008	Prime Minister	Approving the planning for exploration, exploitation, and use of minerals for cement in Vietnam until 2020
9	84/TB-VPCP	16/03/2009	Gov. Office	Conclusion of Deputy PM Nguyen Thien Nhan regarding Trang An tourism environment impacts caused by cement factories
10	657/UBND-VP4	17/07/2009	Ninh Binh PPC	Contributing opinions on the General Construction Planning project of Trang An Tourism Area
11	51/TB-VPCP	24/02/2010	Gov. Office	Conclusion of Deputy PM Hoang Trung Hai regarding the report on General Construction Planning of Trang An Tourism Area

No	Document No	Date	Issuing Authority	Content Summary
12	30/BC-UBND	25/03/2010	Ninh Binh PPC	Report on mineral activities and environmental protection of Duyen Ha & He Duong cement factories
13	998/BTNMT-DCKS	29/03/2010	MONRE	Resolving dossiers for limestone exploration/exploitation for cement within the planning area of Trang An
14	2391/VPCP-KTN	12/04/2010	Gov. Office	Granting licenses for limestone exploration/exploitation for cement projects in Ninh Binh
15	1065/QĐ-TTg	09/07/2010	Prime Minister	Approving adjustments and supplements to the planning for exploration/exploitation of minerals for cement until 2020
16	1728/BXD-KTQH	11/09/2010	Ministry of Construction	Regarding the establishment of General Construction Planning for Trang An Tourism Area
17	844/UBND-VP4	10/12/2010	Ninh Binh PPC	Regarding the establishment of General Construction Planning for Trang An Tourism Area
18	6773/VPCP-KGVX	27/09/2011	Gov. Office	Regarding the preparation of the dossier to submit to UNESCO for World Heritage recognition
19	548/QĐ-TTg	10/05/2012	Prime Minister	Decision on ranking the site as a Special National Relic/Monument
20	1382/TTg-KGVX	13/09/2012	Prime Minister	Regarding the submission of the Trang An Landscape Complex dossier to UNESCO
21	163/BC-UBND	06/12/2012	Ninh Binh PPC	Report on zoning for protection of Trang An Landscape Complex
22	109/BC-BXD	28/12/2012	Ministry of Construction	Inspection and review of cement mineral planning regarding issues related to Trang An Landscape Complex
23	201/QĐ-TTg	23/01/2013	Prime Minister	Approving the Master Plan for Tourism Development of Vietnam until 2020, vision to 2030
24	225/TB-VPCP	01/07/2013	Gov. Office	Conclusion of PM Nguyen Tan Dung on Trang An Complex Planning and Cement Raw Material Zone Planning
25	106/UBND-VP9	25/07/2013	Ninh Binh PPC	Requesting adjustment of the Trang An Landscape Complex buffer zone boundary in the UNESCO submission dossier
26	1651/BXD-VLXD	09/08/2013	Ministry of Construction	Regarding the plan to adjust the buffer zone of Trang An Landscape Complex
27	3033/BVHTTDL-DSVH	20/08/2013	MOCST	Reporting on the plan to adjust the buffer zone of Trang An Landscape Complex for UNESCO submission
28	456/UBND-VP4	29/08/2013	Ninh Binh PPC	Regarding the construction planning of Trang An Landscape Complex
29	7915/VPCP-KGVX	21/09/2013	Gov. Office	Adjusting the buffer zone of Trang An Landscape Complex
30	1266/QĐ-TTg	28/07/2014	Prime Minister	Approving the General Urban Planning of Ninh Binh until 2030, vision to 2050
31	9635/VPCP-KTN	02/12/2014	Gov. Office	Transferring the task of presiding over the formulation of the Master Plan for Trang An Landscape Complex

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Author contributions

Tran Nhat Lam Duyen: Methodology, Conceptualization, Writing – review & editing. Tran Thuc Han: Investigation. Nguyen Van Hieu: Investigation. Hoang Ha Anh: Writing – review & editing, Writing – original draft, Methodology. Nguyen Cam Chi: Investigation. Nguyen Dinh Tien: Data analysis. Nguyen Thi Tuyet Trinh: Conceptualization.

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Data availability

Data are available upon request from the corresponding author.

Declarations

Ethical approval

All recruitment and consent procedures were reviewed and approved by the Ethics Committee of the VNU School of Interdisciplinary Sciences and Arts, Vietnam National University. The procedures used in this study adhere to the tenets of the Declaration of Helsinki.

Consent to participate

Verbal informed consent was obtained prior to the interview.

Competing interests

The authors declare no competing interests.

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